

ORSHA, Belarus

WMO: 267630

Lat: 54.503N

Lon: 30.444E

Elev: 185

StdP: 99.12

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.3	-18.9	-24.5	0.4	-22.1	-21.3	0.6	-18.3	8.9	-1.7	8.2	-3.1	1.3	0	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	28.8	19.8	26.8	18.7	25.1	18.1	20.9	26.6	19.9	25.3	18.9	23.6	3.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	14.0	23.7	17.9	13.2	22.5	17.1	12.5	21.5	61.2	26.6	57.7	25.4	54.5	23.4	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.1	6.3	-26.5	31.2	2.6	2.3	-28.4	32.9	-29.9	34.2	-31.4	35.5	-33.3	37.1		
(5)			-26.6	22.4	2.6	1.5	-28.5	23.5	-30.0	24.4	-31.5	25.2	-33.4	26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-5.6	-5.3	-0.6	7.1	12.5	16.1	18.6	17.1	11.7	6.0	0.8	-3.6		
	DBStd	9.76	6.25	6.57	4.78	4.33	4.29	3.35	3.10	3.38	3.61	4.04	4.86	5.95		
	HDD10.0	2229	485	429	328	109	23	1	0	0	24	135	276	421		
	HDD18.3	4486	743	662	586	338	186	84	35	64	201	383	525	679		
	CDD10.0	877	0	0	1	21	101	183	266	221	74	11	0	0		
	CDD18.3	91	0	0	0	0	6	16	43	26	1	0	0	0		
	CDH23.3	809	0	0	0	3	53	121	376	244	13	0	0	0		
	CDH26.7	188	0	0	0	0	7	13	94	74	1	0	0	0		
(14) Wind	WSAvg	2.9	3.2	3.2	3.1	2.9	2.8	2.6	2.2	2.3	2.5	3.0	3.3	3.3		
(15) Precipitation	PrecAvg	658	41	39	37	40	68	74	82	71	60	59	48	39		
	PrecMax	865	93	69	81	98	126	168	200	236	136	143	108	71		
	PrecMin	489	8	18	8	12	17	28	7	3	14	8	2	15		
	PrecStd	99	18	13	17	21	30	31	45	45	34	35	23	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.8	6.2	13.6	22.7	27.7	28.7	31.5	33.4	25.2	19.5	11.7	8.4		
		MCWB	4.8	4.4	8.3	12.7	19.0	19.0	20.7	20.7	16.8	12.9	10.7	7.3		
	2%	DB	3.6	4.0	10.1	19.7	24.7	26.6	29.3	28.0	22.2	15.3	9.7	6.3		
		MCWB	2.7	2.8	5.9	11.8	17.6	18.7	20.5	18.8	15.8	12.5	8.6	5.5		
	5%	DB	2.4	2.7	7.5	17.0	22.3	24.9	27.2	25.8	19.6	13.3	8.0	4.5		
		MCWB	1.7	1.8	4.4	10.6	15.2	18.1	19.6	18.0	14.3	11.1	6.9	3.9		
	10%	DB	1.3	1.7	5.4	14.7	20.3	22.9	25.3	23.7	17.8	11.7	6.8	2.8		
		MCWB	0.7	1.0	3.3	9.3	14.1	17.0	18.6	17.1	13.3	10.0	6.0	2.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	4.7	8.8	14.2	20.3	21.1	22.8	21.6	17.7	14.6	10.9	7.6		
		MCDB	5.6	5.7	12.7	20.8	26.1	25.7	29.0	29.6	23.1	17.6	11.6	8.2		
	2%	WB	2.9	2.9	6.4	12.4	18.0	19.9	21.5	20.2	16.1	12.9	8.8	5.5		
		MCDB	3.5	3.6	9.2	17.7	23.5	25.1	27.3	26.6	20.3	14.6	9.5	6.0		
	5%	WB	1.7	2.0	4.7	11.2	16.5	18.9	20.2	19.2	15.0	11.3	7.1	3.9		
		MCDB	2.3	2.6	6.8	15.8	21.3	23.5	25.5	24.1	18.5	12.9	7.8	4.5		
	10%	WB	0.8	1.1	3.5	10.0	14.9	17.8	19.2	18.0	13.9	10.1	6.0	2.1		
		MCDB	1.3	1.6	5.3	14.0	18.8	21.8	24.0	21.8	16.8	11.7	6.7	2.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	5.4	6.5	9.1	9.8	9.3	9.9	10.0	9.2	6.3	3.8	3.9		
		MCDBR	3.4	4.1	9.2	13.4	13.0	12.0	12.5	13.7	12.3	8.9	4.9	3.7		
	5% WB	MCWBR	3.4	3.6	5.9	6.9	6.7	5.8	5.5	5.7	6.7	6.1	4.2	3.4		
		MCWBR	3.3	3.8	8.0	11.4	11.7	10.3	11.3	10.8	10.5	7.7	4.4	3.7		
(40) Clear-Sky Solar Irradiance	taub		0.287	0.322	0.351	0.408	0.396	0.378	0.406	0.416	0.373	0.349	0.328	0.290		
		taud	2.135	2.086	2.124	2.171	2.287	2.360	2.299	2.264	2.351	2.382	2.247	2.161		
	Ebn at Noon		634	736	809	813	849	869	834	798	788	719	584	544		
		Edh at Noon	65	97	117	130	122	115	120	117	95	73	60	52		
(44) All-Sky Solar Radiation	RadAvg		0.66	1.56	2.96	4.00	5.24	5.71	5.47	4.43	2.94	1.43	0.52	0.38		
	RadStd		0.15	0.28	0.44	0.47	0.45	0.52	0.52	0.47	0.24	0.29	0.10	0.11		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+1.07	N/A	N/A	N/A	N/A	
(47) Regional (9 neighbors)		+0.59	N/A	N/A	N/A	N/A	N/A	N/A	-174	+98	N/A	

Nomenclature: See separate page